DATA VISUALIZATION LAB PROJECT

The Geography of the Attack

Report (Deliverable 4)

Francesca Michieletto (ID: 255371) · Data Visualization Lab project

PROJECT OVERVIEW

The project:

How do attack zone, attack tempo, and blocking pressure relate to attack efficiency?

This project analyzes attacking performance in the Italian Women's Serie A2 using real DataVolley scouting data.

The analysis focuses on Brescia and Talmassons, the two teams promoted to Serie A1 during the 2025/26 season. Although they reached promotion in different ways (playoffs for Brescia and direct promotion for Talmassons), they finished the regular season only one point apart, making them a suitable comparison.
The dataset consists of scouting files provided by the teams' coaching staff.

63 (36+29)

matches scouted
(Brescia + Talmassons)

16,301

attacks analyzed
(Brescia + Talmassons)

2

head-to-head matches
shared by both teams

FOR READERS NEW TO THE SPORT

Volleyball basics: front row vs. back row

Front row (Zones 2, 3, 4)

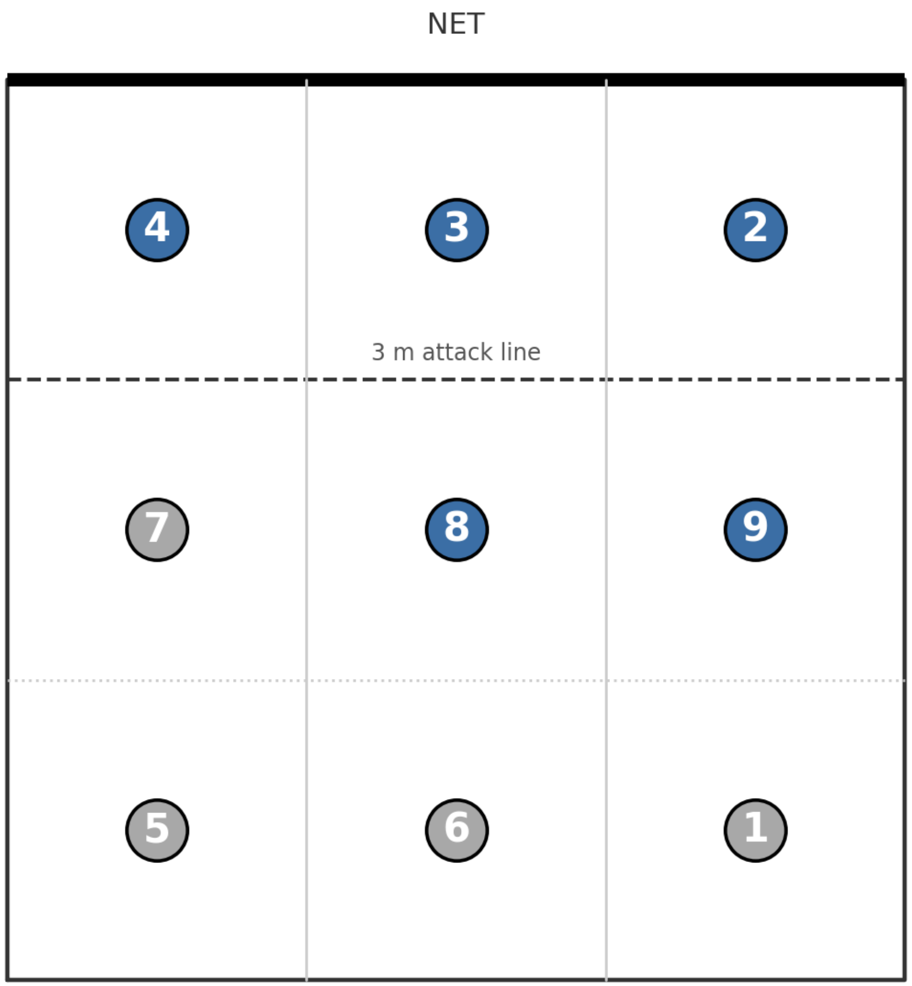
The three players closest to the net. They can attack and block at the net

Back row (Zones 8, 9)

The three players are behind the 3-meter line. They can attack only if they jump from behind this line.

Why this matters

Front-row and back-row attacks are used in different tactical situations. Separating them makes it easier to compare attacking patterns and to understand how each team distributes its attacks.



FROM IDEA TO PROTOTYPE

Project timeline

1

Proposal
Project idea and research questions
Initial visualization design

2

Data parsing
Parsed all 63 DataVolley files with pydatavolley
Recovered attack tempo

3

Exploratory analysis
Dataset validation and EDA
Technical Report
Investigated the num_blockers field

4

First Bokeh prototype
Initial interactive visualization

5

Compatibility fixes
Resolved package compatibility issues
Notebook runs in Google Colab

6

Second Bokeh prototype
Improved layout and interaction

7

Revision
Updated figures
Replaced several visualizations in the Technical Report

8

Final deliverables
Final Google Colab notebook
Standalone HTML application

Results validated against the Technical Report

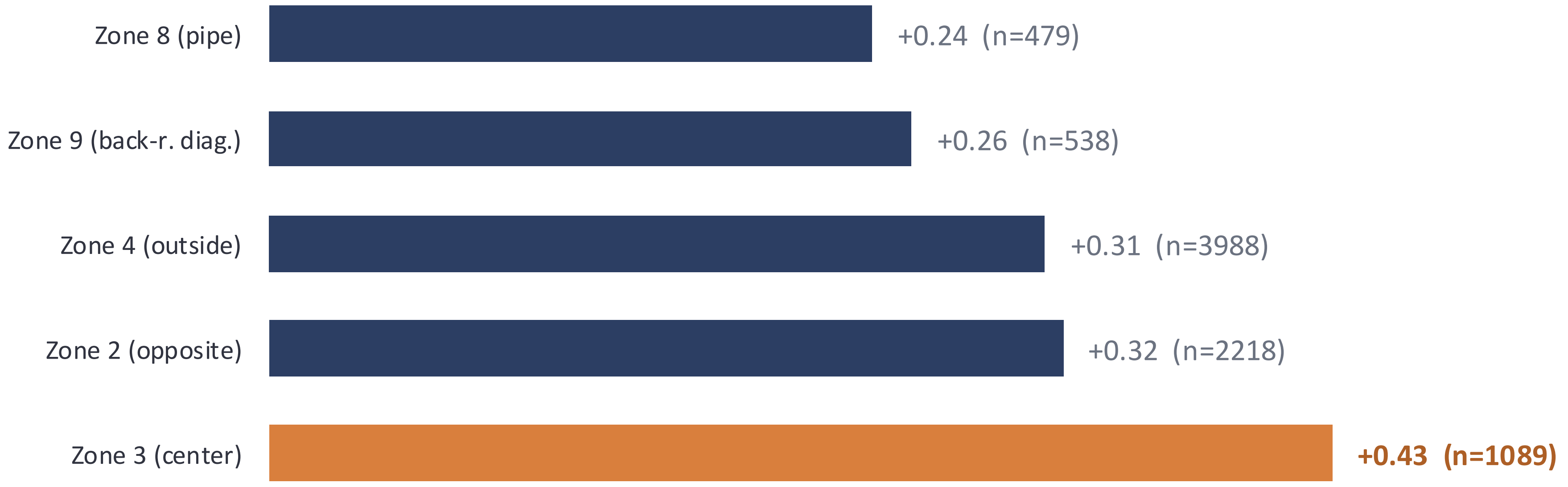
PART ONE

Results

Key findings from the exploratory analysis

Attack efficiency by court zone

The attacks from both teams are ranked by net efficiency (kill rate − error rate). Middle attacks (Zone 3) are the most effective, while back-row attacks from Zone 8 show the lowest efficiency.



Zone 3 has the highest net efficiency (+0.43), while Zone 8 has the lowest (+0.24). This result is consistent with the higher efficiency generally associated with quick middle attacks.

Brescia vs. Talmassons

The scouting profiles of the two teams are almost identical (45.0% positive evaluations for both), supporting a direct comparison of their attacking patterns.

Common pattern

Both teams perform best on quick and medium-tempo attacks through the middle and less effectively on high-ball back-row attacks.

Main difference

The largest difference is observed for high-ball attacks to Zone 2. Since both teams have more than 250 attacks in this category, the difference is supported by a sufficiently large sample.

Brescia	Talmassons
0.20 net eff. (n=320)	0.34 net eff. (n=256)

The largest difference in the dataset. The sample size is large enough for both teams to support the comparison.

PART TWO

What changed and why

How the project changed from the original proposal and why

WHAT CHANGED AND WHY - PROPOSAL VS. FINAL VISUALIZATION

Changes from the original proposal

The final visualization follows the original proposal but includes a few changes that emerged during the implementation and data exploration.

Proposed	Delivered	Why
Court diagram with continuous color scale	Real-proportioned court with discrete efficiency levels	Discrete levels are easier to compare and the court proportions match the real playing area
One combined filter for team + tempo + number of blockers	Separate CDSView + GroupFilter panels, one per dimension	The three views can be compared side by side without changing filters
“Number of blockers” as a filter	Hole block shown as a separate category	The fourth value corresponds to the DataVolley hole block category, not to four blockers
“Pipe” listed as an attack tempo	Pipe identified as Zone 8	Corrected after checking the DataVolley coding

How the dataset evolved during the project

Some variables proposed at the beginning of the project were revised after exploring the data and defining the final scope of the analysis.

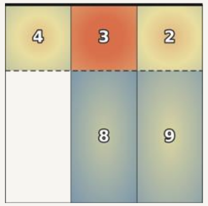
Proposed	Delivered	Why
Attack tempo simplified into named categories (quick, medium, high-ball, pipe)	Five attack-tempo categories (Quick, Super/medium, High ball, Medium, Second ball/other)	Keeping the original categories preserves differences between attack tempos
Filters by setter, rotation, current score, and opponent identity	Only team, zone, tempo, and blocker count are filterable	The project focused on the variables directly related to the research question
Attack side (right/left/middle approach), read from the attack-combination table	Extracted while parsing, but not visualized	It did not provide additional information beyond zone and tempo
Phase of play: first-ball attack vs. transition attack	Season phase (Regular Season, Playoff Pool, Playoffs, Coppa Italia)	Building rally phases would have required additional rally reconstruction that was outside the project scope

Difference 1 — Court diagram

Proposed

Continuous color scale

Single filter for team, tempo and blockers



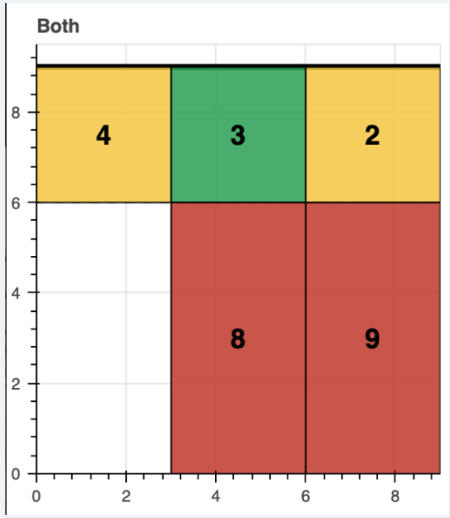
Concept sketch

Delivered

Discrete efficiency levels

Real-proportioned court

Separate CDSView + GroupFilter panels



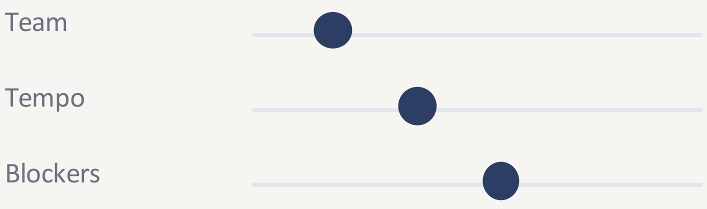
Why: Discrete efficiency levels make the differences easier to compare. Real court proportions provide a more intuitive representation of the attack zones.

Implemented with vbar() using explicit top and bottom coordinates.

Difference 2 — Comparing the teams

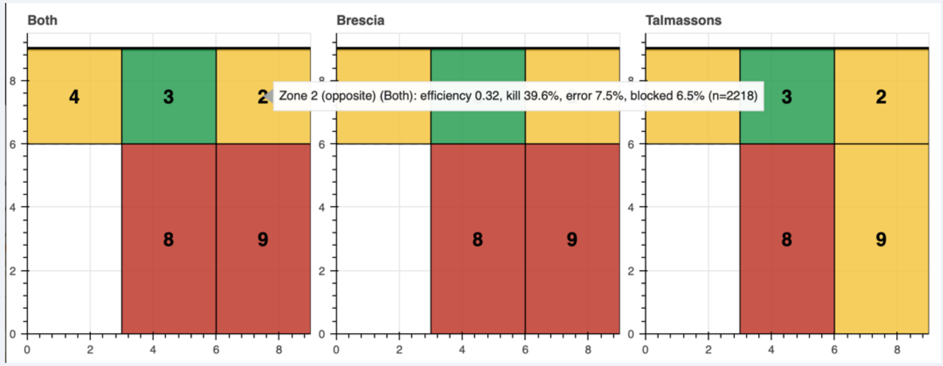
Proposed

Single interactive control panel
- Team
- Attack tempo
- Number of blockers



Delivered

Three linked panels:
- Both teams
- Brescia
- Talmassons



Why: the three panels can be compared directly without changing filters.

Difference 3 — The blocker field

During the exploratory analysis

pydatavolley extracts the num_blockers field directly from the DataVolley action code. Values from 0 to 3 represent the number of opposing blockers.

What changed

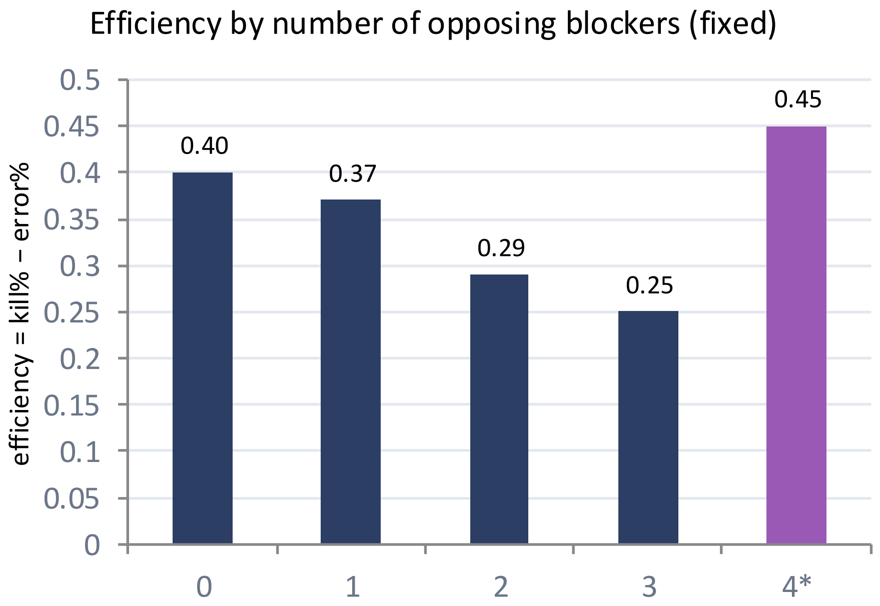
An additional value (4) also appears. At first it looked like an impossible blocker count, so it was investigated before being included in the analysis.

Further documentation showed that 4 is the official DataVolley Hole block category. It describes an attack that passes through a gap in the opposing block, rather than an attack facing four blockers.

Final Decision

The category was kept as a separate value (4*) throughout the analysis and in the Bokeh application.

This verification allowed the blocker field to be used correctly in the final visualization



WHAT CHANGED AND WHY – OTHER CHANGES

Other changes

Terminology

“Pipe” was initially treated as an attack tempo. During data validation, it was correctly identified as Zone 8, representing a back-row attack rather than a tempo.

Future work

PCA on attacking profiles was proposed as an optional extension for this final project.

Additional metric

Block rate (the “/” evaluation code) was added to the hover information, providing additional context for attack outcomes.

PART THREE

Challenges encountered

The main challenges faced during the development of the project

The main challenges

1

Working independently

The project was developed without attending the laboratory sessions, requiring independent study of the course material and Bokeh documentation.

2

Understanding the DataVolley format

Recovering and validating the information needed from the .dvw files.

3

Building the interactive application

Developing and testing the Bokeh dashboard while dealing with package compatibility and implementation issues.

Working independently

Challenge

The project was developed without attending the lessons.

How I approached it

- Studied the course material independently
- Used the Bokeh documentation and the examples discussed during the course
- Built the application incrementally, starting from simple plots before integrating them into a single dashboard
- Discussed the DataVolley scouting conventions with the team's scout analyst to correctly interpret the raw data
- Used AI tools as technical support for debugging, code review and English proofreading (Claude and Grammarly)

Outcome

The complete project, including the notebook, Technical report and interactive Bokeh application, is available on Github:

Understanding the DataVolley format

Challenge

The .dvw files store scouting actions as compact action codes. Although pydatavolley decodes most of the information, attack tempo is not available as a structured field and some variables needed to be validated before the analysis.

Solution

The [3ATTACKCOMBINATION] table was read directly to recover attack tempo and the decoded fields were checked against the original scouting data. Guidance from the team's scout analyst also helped interpret the scouting conventions used in the files.

Outcome

A complete attack-level dataset was obtained with every attack matched to a valid attack zone and attack tempo.

100%

Attack actions matched to a valid attack zone and attack tempo.

63

36 Brescia – 29 Talmassons
Including 2 head-to-head matches

Building the interactive application

Challenge

The final deliverable is an interactive Bokeh application integrating multiple linked visualizations. During development, package compatibility issues also prevented the notebook from running correctly.

How I approached it

- Built and tested each visualization separately before integrating them into the final application
- Investigated and fixed compatibility issues between pydatavolley, NumPy 2.0 and Google Colab
- Validated the notebook against the Technical Report before generating the standalone HTML

Outcome

The final application includes linked interactive visualizations, runs in Google Colab and is also available as a standalone HTML file

np.Nan

Pydatavolley still uses np.Nan which was removed in NumPy 2.0

np.where

NumPy 2.0 introduced stricter type promotion, breaking one of the pydatavolley parsing steps

Double-patch protection

Implemented a compatibility patch while preventing double-patching when rerunning the notebook

CONCLUSIONS

Final remarks

- A visualization is only as reliable as the data behind it.

- Exploring the data changed several design decisions from the original proposal.

- Combining domain knowledge with data visualization led to a more meaningful final application.

CONCLUSIONS - CRITICAL INTERPRETATION

Reading the results critically

The negative correlation may not be purely tactical

Teams with higher efficiency often finish sets more quickly, resulting in fewer attacks. Part of the correlation may therefore reflect match dynamics rather than attacking strategy alone.

The “4” category is not a blocker count

During the project, the fourth value was identified as the official DataVolley Hole block category. This changed the interpretation of the variable and confirmed that it should be treated as a separate category throughout the analysis.

References & Links

DataVolley scouting data (private dataset provided by the teams' coaching staff)
OpenVolley – pydatavolley (Python library for parsing DataVolley files)
OpenVolley – datavolley documentation (reference for DataVolley codes and file structure)
Bokeh documentation

GitHub repository: